Discovering the Memory Structure of FX Volatility: A LASSO-Based Kernel Identification Approach

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1 Motivation

1.1 Volatility: The True Currency of Finance

Volatility is the true currency of the financial industry. While prices capture the level of asset values, volatility captures their risk, and risk, not return, is what traders, risk managers, and regulators truly trade and manage. Unlike prices that follow near-random walks, volatility exhibits persistent, predictable patterns. This predictability makes volatility modeling one of the longest-standing and most consequential problems in quantitative finance.

The discovery of the Black-Scholes-Merton option pricing formula in 1973 revolutionized finance by providing, for the first time, a rigorous framework to value derivatives. Within two years of publication, the Chicago Board Options Exchange launched standardized options trading. By the 1980s, derivatives had evolved from niche instruments to core components of corporate risk management and investment strategy.

However, the original Black-Scholes model contained a critical simplifying assumption: constant volatility, an assumption convenient for mathematical tractability but increasingly recognized as unrealistic. Throughout the 1980s, traders observed that implied volatilities extracted from market prices varied systematically with strike price and maturity, creating the famous "volatility smile" and "term structure" that the constant volatility assumption could not explain. Markets were pricing options as if volatility itself was stochastic and time-varying, but the mathematical models had not yet caught up.

The fragility of the constant volatility assumption became catastrophically apparent on October 19, 1987, Black Monday, when the Dow Jones Industrial Average crashed 22.6% in a single day, the largest one-day percentage decline in stock market history. The lesson was stark: accurate volatility modeling is not merely an academic nicety but a practical necessity for market stability. Since 1987, understanding and predicting volatility has become central to pricing, hedging, and risk management of portfolios of derivatives.

1.2 Competing Paradigms: Stochastic versus Endogenous Volatility

The recognition that volatility is time-varying and stochastic spawned two fundamentally different modeling approaches over the subsequent decades.

The first paradigm treats volatility as driven by exogenous stochastic factors. Stochastic volatility models, pioneered by Hull and White (1987) and refined by Heston (1993), introduce separate random processes for volatility itself.

The second paradigm models volatility as endogenous, determined by observable market variables rather than hidden stochastic factors. The ARCH (Autoregressive Conditional Heteroskedasticity) model introduced by Engle (1982) and its generalization GARCH by Bollerslev (1986) represent the canonical examples. These models specify volatility as a function of past returns and past volatility realizations, making volatility entirely path-dependent on observable history.

1.3 The Paradigm Shift: Path-Dependent Volatility

The work by Julien Guyon and Jordan Lekeufack on Path-Dependent Volatility represents a paradigm-shifting approach that fundamentally reframes this debate. Published in Quantitative Finance in 2023 and earning Guyon the 2025 Risk Magazine Quant of the Year award, their research demonstrates that volatility can be explained almost entirely by endogenous factors, specifically past returns, with minimal need for exogenous stochastic drivers.

The PDV framework postulates that instantaneous volatility is a deterministic function of two path-dependent features constructed from historical returns: a trend feature capturing directional momentum and asymmetric effects, and a volatility feature capturing persistence and clustering.

The remarkable discovery was that a simple linear four-parameter PDV model could explain up to 90% of the variance in implied volatility (VIX) for the S&P 500, with test R^2 values of 0.855. This performance vastly exceeded GARCH models, stochastic volatility models including Heston, rough volatility models, and all competing approaches.

1.4 Research Objectives

While Guyon's results for equity index volatility are groundbreaking, foreign exchange markets remain unexplored territory for PDV analysis. FX markets exhibit fundamentally different characteristics that may produce distinct memory structures.

FX markets trade continuously 24 hours per day across global time zones, contrasting with equity indices that have defined opening and closing times. This continuous trading eliminates overnight gaps and may create different memory patterns. Volatility levels in FX are substantially lower (5-15% annualized for major pairs) compared to equity indices (15-30%), suggesting potentially different underlying dynamics.

These differences motivate a critical empirical question: Does FX volatility memory follow the same power-law structure discovered for equities, or do FX-specific dynamics produce distinct kernel characteristics?

This research applies the LASSO-based kernel discovery methodology to EUR/USD spot rates spanning 2012-2021, with three primary objectives:

1. We identify the functional form of EUR/USD volatility kernels to determine whether they follow power-law decay similar to equity indices, exponential decay as traditional models assume, or some alternative functional form. The pattern of LASSO-selected decay rates will reveal whether FX volatility has a singular memory time scale or multi-scale memory spanning hours to months.

2. We conduct comparative analysis against Guyon's equity findings to identify similarities and differences in memory structure.

3. We evaluate predictive performance to determine whether data-driven kernel discovery improves volatility forecasting.

The contributions extend beyond replication of Guyon's methodology in a different asset class. FX markets represent the world's largest and most liquid financial market, with daily turnover exceeding 7.5 trillion dollars.

2 Theoretical Framework: Path-Dependent Volatility

2.1 The Path-Dependent Volatility Hypothesis

The path-dependent volatility (PDV) framework postulates that current volatility is a deterministic function of historical return patterns rather than an exogenous stochastic process. Formally, instantaneous volatility σ_t at time t is expressed as:

$$\sigma_t = f(R_{1,t}, \Sigma_t) \tag{1}$$

where $R_{1,t}$ captures trend, and Σ_t captures volatility persistence and clustering. This formulation challenges traditional stochastic volatility models by asserting that volatility dynamics are primarily endogenous, driven by observable market history rather than unobservable latent factors.

2.2 Trend Features: Capturing the Leverage Effect

Trend features capture recent directional movements in asset prices to model the leverage effect—the empirical observation that volatility rises when prices fall. The primary trend feature is defined as:

$$R_{1,t} := \sum_{t_i \le t} K_1(t - t_i) \cdot r_{t_i} \tag{2}$$

where $r_{t_i} := \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}}$ denotes the return between consecutive observation times, and $K_1 : \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ is a convolution kernel that assigns weights to past returns based on their lag $t - t_i$.

Kernel properties. The kernel K_1 determines the memory structure of volatility's response to directional moves.

2.3 Volatility Features: Capturing Clustering and Persistence

Volatility features capture the autocorrelation structure of volatility itself, implementing the empirical regularity that periods of high volatility cluster together. The canonical volatility feature is:

$$R_{2,t} := \sum_{t_i \le t} K_2(t - t_i) \cdot r_{t_i}^2 \tag{3}$$

where $K_2 : \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ is a second convolution kernel, typically distinct from K_1, that weights past squared returns (a proxy for realized volatility).

Volatility-scale transformation. For dimensional consistency with instantaneous volatility, $R_{2,t}$ is transformed to volatility scale:

$$\Sigma_t := \sqrt{R_{2,t}} \tag{4}$$

This square root transformation ensures Σ_t has units of volatility (percentage per square root time) rather than variance, and also compresses extreme values to improve numerical stability with constraints $\beta_0 > 0$ (positive baseline volatility), $\beta_1 < 0$ (leverage effect: negative returns increase volatility), and $\beta_2 \in (0, 1)$ (volatility persistence with stability).

3 Data Acquisition and Preprocessing

3.1 Data Source and Sampling

We obtain high-frequency foreign exchange data from Dukascopy, a Swiss online bank and brokerage firm based in Geneva. The dataset consists of 1-minute OHLC (Open-High-Low-Close) bars for the EUR/USD currency pair spanning January 1, 2012 to December 31, 2021, providing over 11 years of continuous tick-level market activity.

3.2 Data Cleaning and Quality Control

High-frequency financial data invariably contains artifacts from market microstructure: timestamp duplicates, flat bars during low-activity periods, and missing observations during weekends or system outages. We implement a rigorous cleaning pipeline to ensure data integrity.

Step 1: Missing timestamp identification. We construct a complete 1-minute time grid spanning the data range and identify missing timestamps:

```
full_range = pd.date_range(start=data.index.min(),
                           end=data.index.max(), freq='1min')
missing_timestamps = full_range.difference(data.index)
```

Step 2: Duplicate removal. Duplicate timestamps arise from data feed issues or reporting errors. We remove all but the first occurrence of any duplicated timestamp:

```
to_remove = df_clean['DateTime'].duplicated(keep='first')
df_clean = df_clean.loc[~to_remove]
```

Step 3: Consecutive duplicate and flat bar handling. We identify and mark three types of problematic observations:

1. Consecutive duplicates (backward): Bar i has identical OHLC to bar $i-1$
2. Consecutive duplicates (forward): Bar i has identical OHLC to bar $i+1$
3. Flat bars: Open = High = Low = Close (no price movement)

These patterns indicate either data replication errors or complete absence of trading activity. Rather than deleting these observations (which would disrupt the time series structure), we mark them with an indicator and set OHLC values to NaN:

```
is_duplicate = prev_dup | next_dup | flat_bar
df_clean.loc[is_duplicate, ['Open', 'High', 'Low', 'Close']] = np.nan
df_clean.loc[is_duplicate, 'indicator'] = 1
```

This approach preserves the temporal structure while ensuring contaminated bars don't distort volatility estimates.

Cleaning results:

- Original observations: 3,712,747
- Missing timestamps: 1,545,772

- Duplicate timestamps removed: 180
- Flat bars/consecutive duplicates marked: 57,130
- Final clean dataset: 5,258,339

4 Feature Engineering and Model Specification

4.1 Exponential Kernel Basis Functions

Following Guyon & Lekeufack (2023), we construct path-dependent features using exponential kernels with varying decay rates to discover the underlying memory structure of EUR/USD volatility. The approach tests whether volatility kernels can be approximated as linear combinations of exponential functions:

$$K(\tau) \approx \sum_{j=1}^{J} \beta_j \lambda_j e^{-\lambda_j \tau} \tag{5}$$

where λ_j are decay parameters controlling memory length, and β_j are coefficients to be estimated via LASSO regression.

Lambda grid specification. We define a lambda grid $\Lambda = \{\lambda_1, \lambda_2, \ldots, \lambda_{19}\}$ spanning three orders of magnitude to capture memory across multiple time scales:

$$\Lambda = \{0.001, 0.002, 0.005, 0.01, 0.02, 0.03, 0.05, 0.07, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7, 1.0, 2.0, 3.0, 5.0, 10.0\} \tag{6}$$

4.2 Feature Construction

For each decay rate $\lambda_j \in \Lambda$, we construct two kernel-weighted features capturing distinct aspects of volatility dynamics:

$$R_{1,t}^{(\lambda)} = \sum_{i=0}^{t} \lambda e^{-\lambda(t-i)} \cdot r_i \tag{7}$$

where $r_i = \log(S_i/S_{i-1}) \times \sqrt{365 \times 12}$ is the annualized log return.

$$R_{2,t}^{(\lambda)} = \sqrt{\sum_{i=0}^{t} \lambda e^{-\lambda(t-i)} \cdot r_i^2} \tag{8}$$

Resulting feature matrix. With 19 lambda values and 2 feature types per lambda, we generate 38 candidate features:

- Features 1-19: $\{R_{1,t}^{(\lambda_1)}, R_{1,t}^{(\lambda_2)}, \ldots, R_{1,t}^{(\lambda_{19})}\}$
- Features 20-38: $\{R_{2,t}^{(\lambda_1)}, R_{2,t}^{(\lambda_2)}, \ldots, R_{2,t}^{(\lambda_{19})}\}$

After removing a 200-period burn-in to allow recursive kernels to stabilize, the final feature matrix has dimensions $31,391 \times 38$.

4.3 Efficient Recursive Implementation

Direct computation of kernel-weighted sums has $O(T^2)$ complexity—prohibitive for $T \approx 31,000$ observations. However, exponential kernels satisfy a recursive property:

$$R_t^{(\lambda)} = e^{-\lambda} \cdot R_{t-1}^{(\lambda)} + r_t \tag{9}$$

$$\begin{aligned} R_t^{(\lambda)} &= \sum_{i=0}^{t} \lambda e^{-\lambda(t-i)} r_i \\ &= \lambda r_t + \lambda \sum_{i=0}^{t-1} e^{-\lambda(t-i)} r_i \\ &= r_t + e^{-\lambda} \sum_{i=0}^{t-1} \lambda e^{-\lambda((t-1)-i)} r_i \\ &= r_t + e^{-\lambda} R_{t-1}^{(\lambda)} \end{aligned} \tag{10}$$

This reduces complexity to $O(T \times K)$ where $K = 19$ lambdas, enabling computation in under 1 second for the full dataset.

4.4 Model Selection: LASSO

The kernel discovery problem presents a fundamental challenge: we have 38 highly correlated candidate features (adjacent lambdas produce correlations > 0.98) competing to explain volatility. OLS regression includes all 38 features with non-zero coefficients, cannot distinguish signal from noise, and is unstable with multicollinearity.

LASSO objective function:

$$\hat{\beta}^{LASSO} = \arg\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{38} \beta_j x_{ij})^2 + \alpha \sum_{j=1}^{38} |\beta_j| \right\} \tag{11}$$

where α controls the regularization strength (larger $\alpha \to$ more sparsity).

4.5 Model Training Procedure

Train-test split. We employ a strict 70-30 time-series split with no shuffling to preserve temporal ordering. This ensures the model never sees future information when predicting past volatility, and that out-of-sample performance is tested on genuinely unseen market regimes.

Feature standardization. Prior to LASSO regression, we standardize features to zero mean and unit variance:

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j} \tag{12}$$

This ensures the L1 penalty treats all features equally regardless of their native scale. Standardization is fit on training data only, then applied to test data to prevent information leakage.

Cross-validation for hyperparameter tuning. We train two LASSO variants:

1. Fixed-alpha LASSO: $\alpha = 0.1$ (arbitrary baseline for comparison)

2. LassoCV: Uses 5-fold time-series cross-validation to select optimal α

This respects temporal ordering: each fold uses past data to predict future data, never the reverse. The optimal α balances training fit against validation performance, selecting the sparsity level that generalizes best.

5 Model Specifications

5.1 Model Overview

To comprehensively evaluate if FX markets exhibit the same path-dependent nature as equity markets, we implement five distinct modeling approaches spanning from a similar framework to Guyon & Lekeufack to further expansions using machine learning. We implement the two LASSO variants: one with fixed alpha ($\alpha = 0.1$) as a baseline, and one with cross-validated alpha (LassoCV) for more robust hyperparameter selection. Additionally, we implement a linear PDV model and extend it further to a Random Forest and XGBoost PDV model. Because the features used in the original PDV model underperform at sharp volatility jumps, two new features were added. The jump indicator flags return observations exceeding 1.75 standard deviations from the trailing mean. The other feature is volatility of volatility which captures when volatility transitions between calm and turbulent states.

5.2 Linear PDV Model

The Linear PDV model implements the core specification from Guyon & Lekeufack (2023) suggesting that instantaneous volatility is a linear function of two path-dependent features:

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t \tag{13}$$

where $R_{1,t}$ is the trend feature and Σ_t is used to capture recent volatility.

5.3 Random Forest PDV

To extend the linear PDV into a nonlinear framework, and see if there is greater performance, a Random Forest regression model was used. The original linear PDV model used power law decay kernels that gave a long memory to past returns. In foreign exchange these older observations are less informative due to shifts in macroeconomics. Therefore, the new idea was to use exponential kernels which have a finite memory in the linear PDV model.

5.4 LASSO / LassoCV

The LASSO methodology and training procedure are detailed in Sections 4.4 and 4.5. In short, both variants use 38 exponential kernel features with L1 regularization for automatic feature selection. The fixed alpha LASSO ($\alpha = 0.1$) provides a baseline whereas the LassoCV uses 5-fold time-series cross-validation to select optimal α.

5.5 XGBoost PDV

XGBoost (Extreme Gradient Boosting) is a scalable tree boosting algorithm that constructs an ensemble of K regression trees, where each tree corrects the residual error of the previous tree:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i) \tag{14}$$

The model minimizes the objective function:

$$\mathcal{L} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \sum_{k=1}^{K} \left[\gamma T_k + \frac{1}{2} \lambda \|w_k\|^2 \right] \tag{15}$$

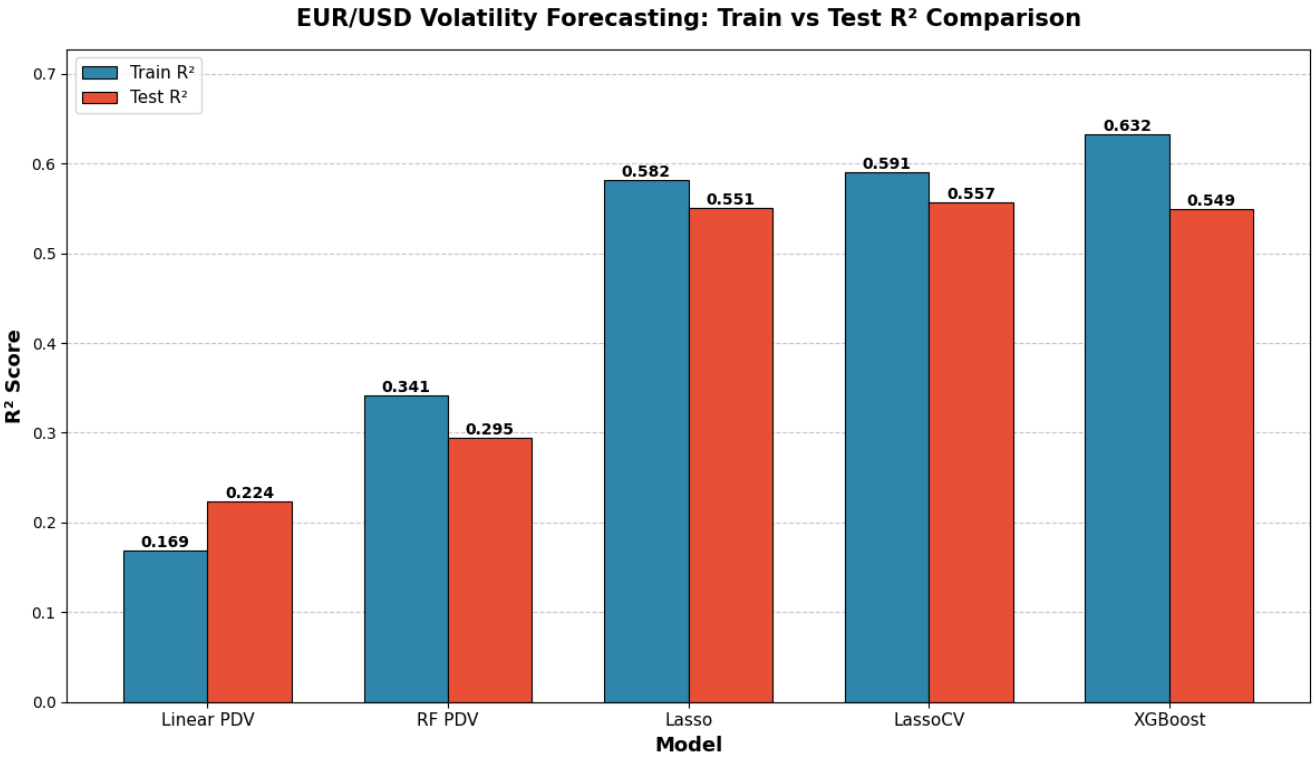
where the first term measures prediction error and the second term penalizes model complexity through both the number of leaves (T_k) and the magnitude of leaf weights (w_k).

We select XGBoost to test whether nonlinear relationships exist in FX volatility that linear models cannot capture. For fair comparison, we train the model on the identical 38 exponential kernel features and use the same 70-30 train-test split. If XGBoost substantially outperforms the linear models, this would suggest FX volatility exhibits nonlinear dynamics that LASSO cannot capture. Conversely, if performance is similar, we can confirm that a linear relationship is sufficient for modeling path-dependent volatility in FX markets. XGBoost is also computationally efficient, training on our $31,000 \times 38$ feature matrix in under 30 seconds with built-in early stopping to prevent overfitting.

6 Evaluation & Analysis

6.1 Model Performance Summary

Figure 1 presents the train and test R^2 scores across all five models. The results reveal a clear distinction: the 38-feature exponential kernel models (LASSO, LassoCV, and XGBoost) substantially outperform the simpler specifications (Linear PDV and Random Forest PDV). LassoCV achieves the highest test R^2 at 55.7%, followed closely by LASSO (55.1%) and XGBoost (54.9%).



The Linear PDV model, which directly applies Guyon & Lekeufack's equity-calibrated power-law kernel parameters, achieves only 22.4% test R^2, demonstrating that kernel parameters optimized for daily S&P 500 data do not generalize to intraday EUR/USD markets. Even when extending the framework to Random Forest with additional engineered features, test performance reaches only 29.5%, confirming that the limited two-feature specification cannot match the multi-scale memory captured by the 19-lambda exponential kernel grid.

6.2 Linear vs. Non-Linear Models

An important finding emerges from comparing the LASSO models and the XGBoost model: despite XGBoost being a predominantly nonlinear model, it performs nearly identically to the LASSO variants. XGBoost achieves a test performance of 54.9% compared to the linear LassoCV model which achieves 55.7%. The XGBoost model also exhibits a larger train-test gap (63.2% vs. 54.9%) compared to LassoCV (59.1% vs. 55.7%), indicating mild overfitting despite regularization through early stopping and tree depth constraints. The fact that XGBoost fails to improve upon the linear baseline strongly suggests that nonlinear patterns either do not exist or are too weak to reliably generalize out-of-sample.

This result confirms Guyon & Lekeufack's finding that nonlinear models do not significantly outperform linear ones for path-dependent volatility. The relationship between historical returns and future volatility is predominantly linear in the exponential kernel features, even in FX markets. This finding provides support for the theoretical simplicity of the PDV framework: volatility responds linearly to weighted combinations of past returns and squared returns across multiple time scales.

6.3 Feature Importance

The LASSO coefficients reveal which exponential decay rates (λ) carry the strongest predictive signals for EUR/USD volatility. With 19 lambda values spanning from 0.001 to 10, the L1 regularization automatically performs feature selection by driving uninformative coefficients to exactly zero. Simply put, LASSO retains the coefficients that matter while eliminating those that don't contribute to the prediction.

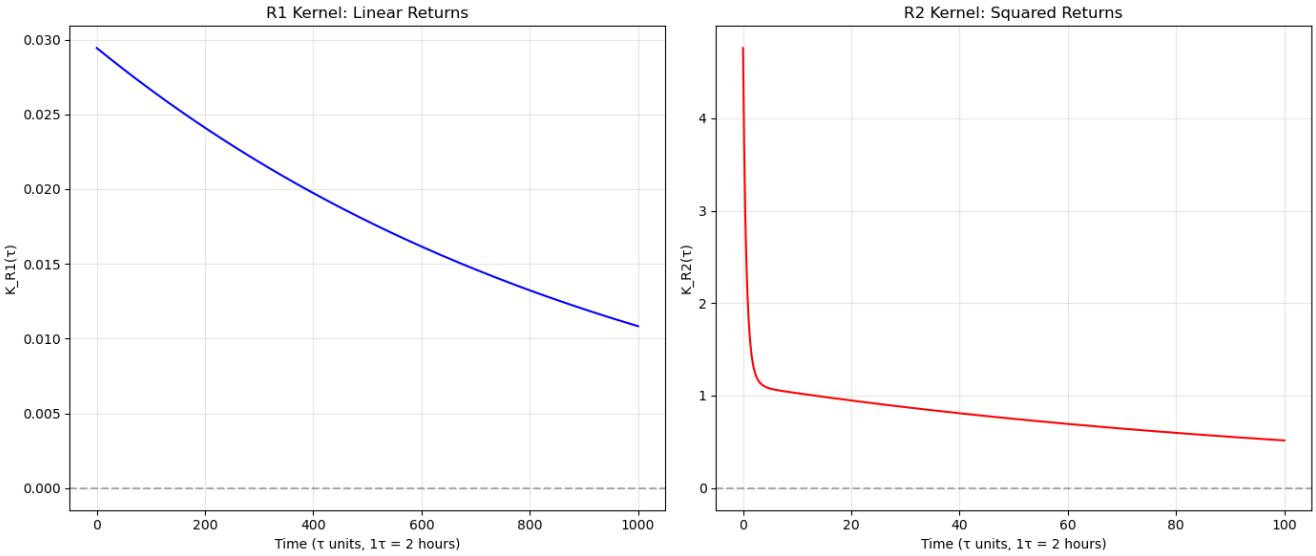
The volatility features (R_2) contribute more to the model than the trend features (R_1).

6.4 Comparison to Results from Guyon and Jordan Lekeufack (2023)

The R_1 kernel (linear returns) showing slower decay in EUR/USD is quite telling. In Guyon and Lekeufack's S&P work, both kernels exhibited similar decay rates, but the FX results show the signed return memory persists much longer—spanning roughly 2000 hours before decaying to $\sim$0.01. This suggests that directional momentum effects in FX have a longer half-life, which makes intuitive sense given that currency markets are driven by macroeconomic trends, central bank policy cycles, and carry trade dynamics that unfold over weeks or months rather than days.

The R_2 kernel preserving the characteristic "fast short-term + slow long-term" structure is reassuring—it confirms that volatility clustering is a universal feature across asset classes, even if the specific time constants differ.

The R^2 of 0.55 is respectable for out-of-sample volatility forecasting, though the diagnostics reveal some areas for improvement. The residual plot shows clear heteroscedasticity—the model underestimates during high-volatility regimes. The Q-Q plot's heavy tails confirm this: the model struggles with extreme events.



6.5 Final Remarks

This project provides an application of the ideas in "Volatility Is (Mostly) Path-Dependent" by Julien Guyon and Jordan Lekeufack to EUR/USD and see if the ideas hold. The paper examined how much of volatility can be explained from past returns using a linear model. Through testing the linear model they presented we discovered that their two features (Σ_t and R_1) are not enough to accurately capture volatility in foreign exchange. Our results show that models using an

exponential kernel with a broad spectrum of decay rates in the feature set lead to outperformance compared to those that incorporated the features presented in the paper. This is because it gives the model a greater understanding of the volatility landscape. Future work would include testing other currencies, and improving the features set to include macroeconomic data as that is what drives foreign exchange markets.

7 Contributions

This research project was a collaborative effort with each team member contributing distinct contribution across different phases of the analysis.

Benjamin Andresen Exploratory data analysis phase, implemented and tested both the Linear PDV and Random Forest PDV models, including the development of custom scikit-learn wrapper classes to ensure consistent interfaces for model training, prediction, and evaluation. engineered the expanded feature set for the Random Forest model.

George Psarros developed the XGBoost PDV model, conducted the comprehensive model comparison and analysis.

Arvinder Singh research paper discovery and review of the path-dependent volatility literature as the foundational framework, data acquisition from Dukascopy, developed the LASSO-based kernel identification methodology, and conducted the kernel decay analysis comparing FX memory structure to equity market findings.

References

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